

- 4 (a) Explain what is meant by an *ideal gas*.

It is a gas that obeys the equation of state $PV=nRT$ for all values of pressure P , volume V and temperature T for a fixed amount n of gas.

[1]

- (b) A fixed mass of ideal gas has a volume of 210 cm^3 at pressure $3.0 \times 10^5 \text{ Pa}$ and a temperature of 35°C .

- (i) State and explain the assumption of the kinetic theory that allows a gas to maintain its temperature.

The assumption is all collisions are elastic.

If the collisions are elastic, total kinetic energy of the gas molecules will be constant

Hence the average kinetic energy will also be constant resulting in the temperature remains constant.

[2]

- (ii) The volume of the gas is then reduced at constant pressure to 140 cm^3 by a moving piston. Determine the final temperature of the gas.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{210}{35 + 273} = \frac{140}{T_2}$$

$$T = 205 \text{ K}$$

temperature of gas = K [2]

- (iii) Calculate the average kinetic energy of a gas molecule at this final temperature.

$$E_k = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(205)$$

$$= 4.24 \times 10^{-21} \text{ J}$$

average kinetic energy = J [2]

- (iv) Using the first law of thermodynamics, explain whether heat is supplied to or released by the gas.

Positive work is done on the gas during compression. Since temperature of the gas decreases, its internal energy also decreases. Therefore the change in internal energy is negative. By the first law of thermodynamics, heat transfer will be negative resulting in heat released by the gas.

[2]

[Total: 9]

